CS261 Airport Simulation - User Manual

Group Members:

Crystal Ng Shu Lu

Saroop Jagdev

Federico Parisella

Julia Chrom

Turki Aldossary

Jainam Kamdar

Group: 12

University of Warwick

Department of Computer Science

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1 Introduction

The goal of this document is to explain how to use the provided application, and interpret the results gathered from the simulation.

1.1 Starting the application

First, ensure that all the required dependencies are installed by either running this command in the terminal for each package:

```
pip install *package_name*
```

or alternatively running this command in the terminal, in the **airport_sim** directory:

```
pip install -r requirements.txt
```

These are all the required packages:

- **Flask** 3.1.2
- **matplotlib** 3.10.8
- **numpy** 2.4.2
- **pytest** 9.0.2
- **Jinja2** 3.1.6
- **Flask-Bootstrap** 5.3.8
- **python-dotenv**

The app can then be run using the command in the terminal, in the **airport_sim** directory:

```
python3 app.py
```

2 Starting a Simulation

To begin a new simulation, click on the blue 'Start New Simulation' button, which will then prompt you to give the inputs to the simulation.

2.1 Simulation Input

This is the information required to start a simulation:

- Departure Runways - runways used by departing planes
- Landing Runways - runways used by landing planes
- Mixed Runways - runways used by either departing or landing planes

(The sum of these 3 must be between 1 - 10)

- Inbound flow - no. of planes/hour landing
- Outbound flow - no. of planes/hour departing

(Both of these between 1 - 100)

- Medical Emergency Probability - a decimal representing chance
- Health Emergency Probability - a decimal representing chance
- Fuel Emergency Probability - a decimal representing chance
- Mechanical Emergency Probability - a decimal representing chance

(All of these are decimals between 0 - 1)

- Weather Closure Probability - a decimal representing chance
- Safety Closure Probability - a decimal representing chance
- Construction Closure Probability - a decimal representing chance
- Maintenance Closure Probability - a decimal representing chance
- Runway Opening probability - a decimal representing chance

(All of these are decimals between 0 - 1)

- Cancellation time - no. of minutes a plane waiting to depart will wait before being cancelled
- Simulation duration - no. of minutes that the simulations will run for

(Simulation duration must be between 10 - 1440)

The simulation is then started by pressing the 'Start Simulation' button at the bottom of the screen. This will generate randomised planes according to the given input parameters.

2.2 Default Probabilities

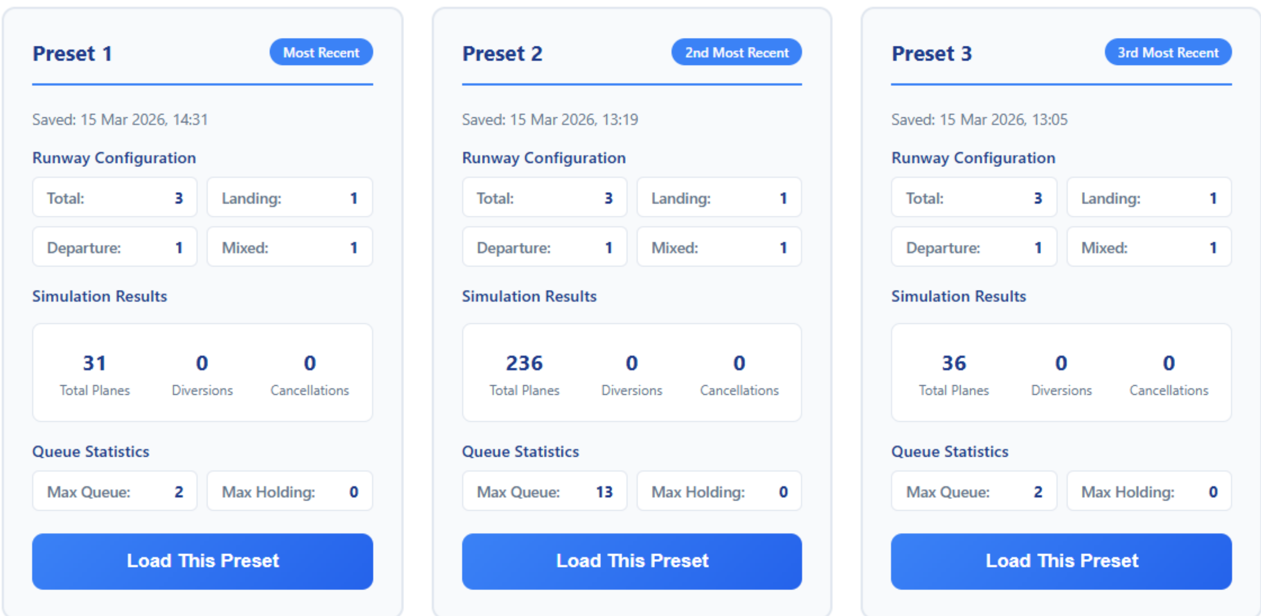
The default values included for the probabilities parameters come from **yearly airport data** gathered from airports around the world, and so can be left as **default** for a **realistic chance** of emergencies occurring.

2.3 Using a Preset

Alternatively you can use an already ran simulation, and alter it slightly to create different results that can then be compared.

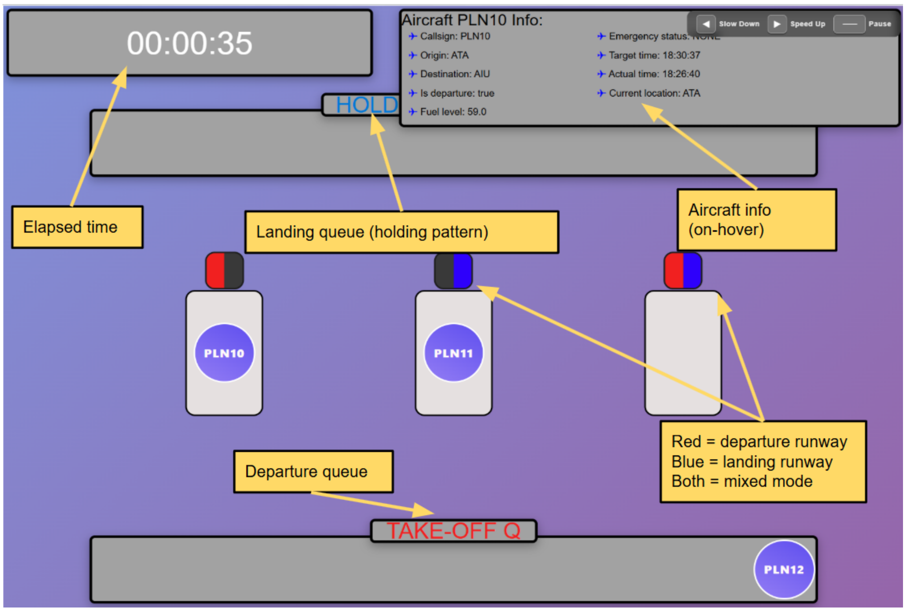
From the main menu, selecting 'Load Preset' will take you to a preset select screen. These consist of the 3 most recent simulations run. By pressing on 'Load This Preset', the normal simulation input screen is displayed if you wish to alter some input data. Then the simulation that runs subsequently will not generate new planes but rather use the ones from the previous simulation preset.

This allows for comparison of different situations under the same scenario conditions.



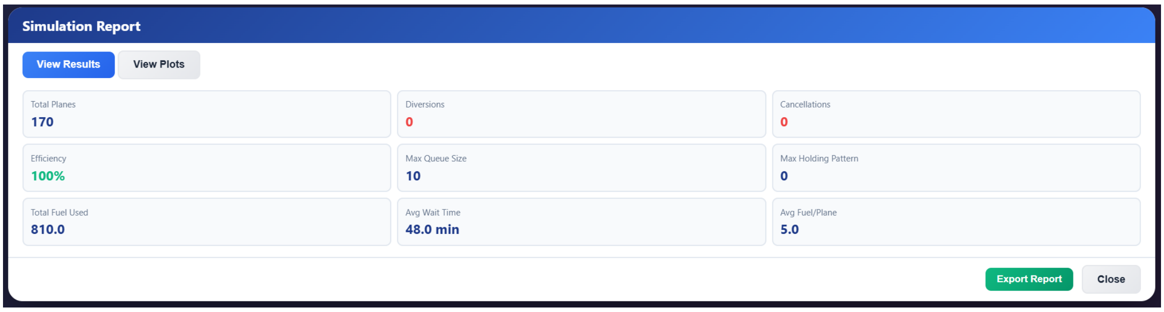
2.4 Interpreting the Simulation Display

This simple diagram explains what each thing represents on the simulation display. You can also **speed-up/slow-down** the simulation by using the **left/right arrow keys** respectively. The simulation may also be **paused** by pressing the **space bar**.



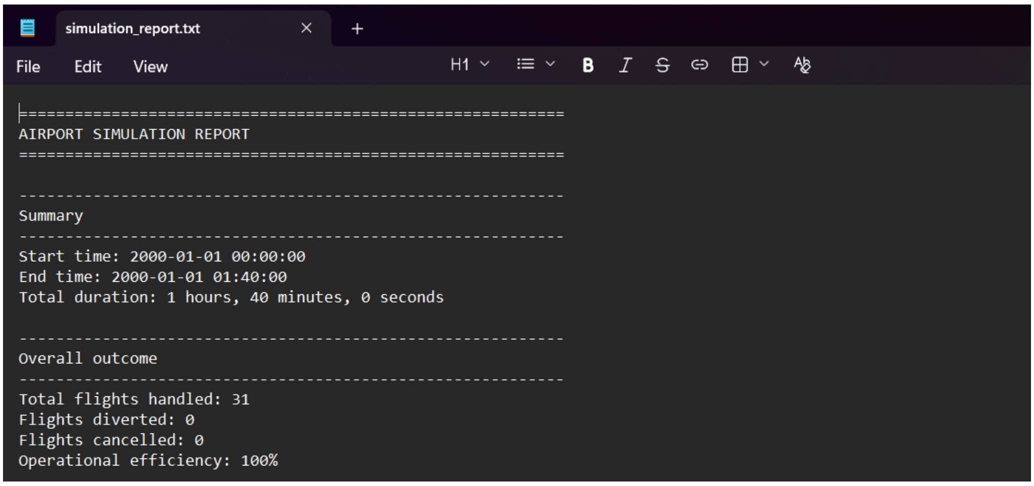
3 Simulation results

When a simulation has terminated, the statistics being tracked will be displayed, with the option to see plots.



3.1 Exporting results

At the bottom right of the results screen there is a 'Export Report' button, which when clicked on, will download the report throught the browser, as a **.txt file**, which will include all the shown parameters + go into more detail for some further calculations.



3.2 Comparing results

Back in the main menu, by clicking the 'View Results' button, you will be taken to a list of all the results from previously run simulations. Here you can view each one individually or select 2 to compare by first checking the tickbox for the chosen results, and then clicking the 'Compare selected' button.